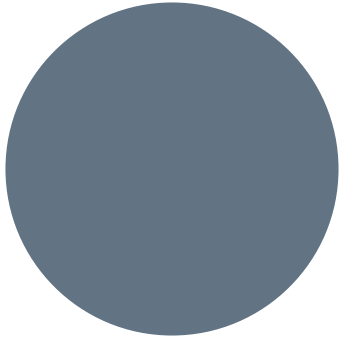
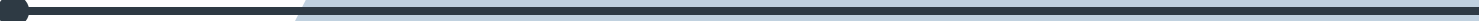




Deepfake Detection Benchmark



EE656-Course Project
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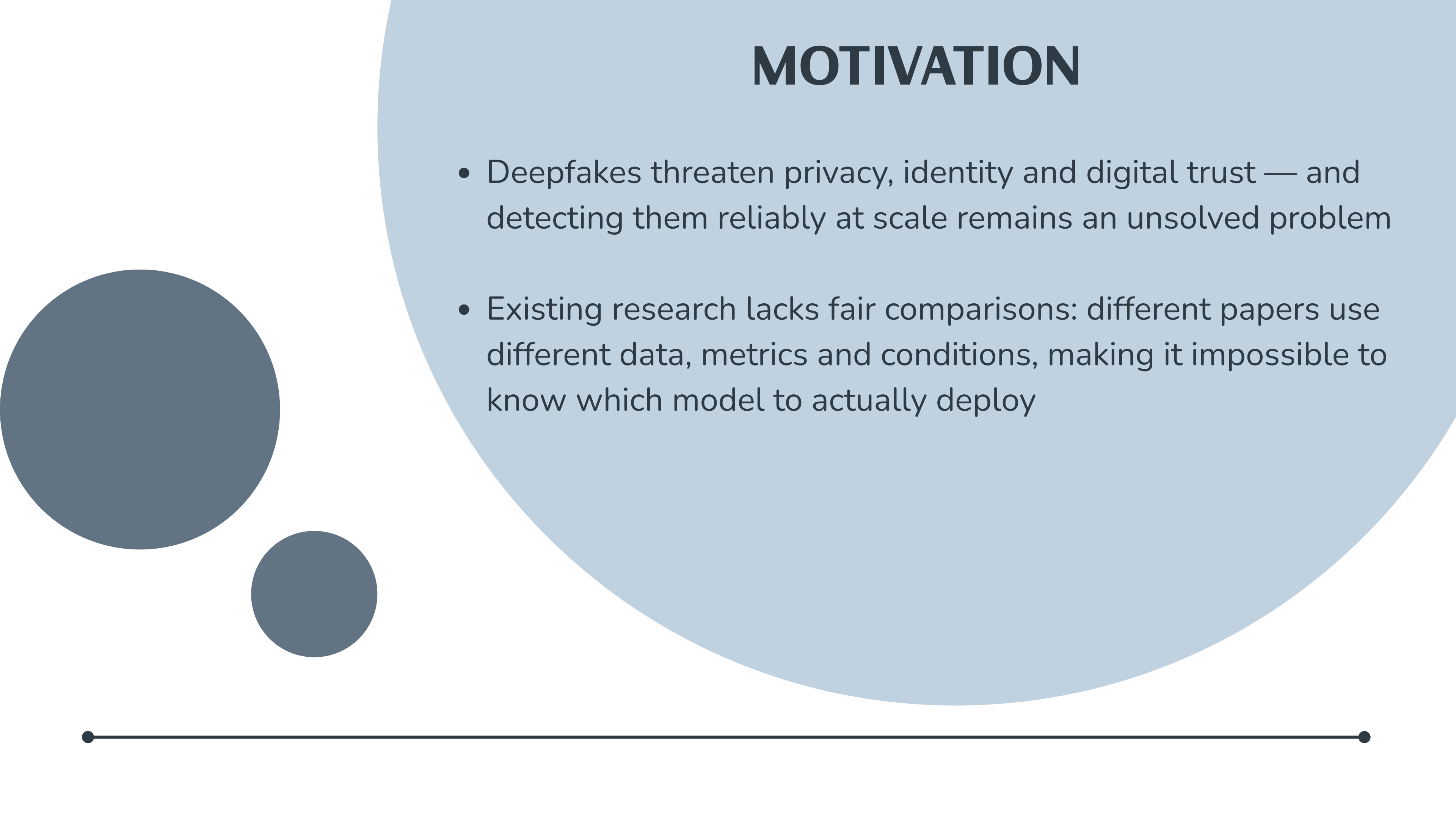
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MOTIVATION

- Deepfakes threaten privacy, identity and digital trust — and detecting them reliably at scale remains an unsolved problem
 - Existing research lacks fair comparisons: different papers use different data, metrics and conditions, making it impossible to know which model to actually deploy
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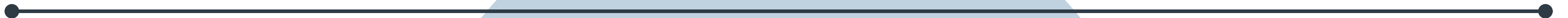
OBJECTIVES

- Train MesoNet-4 and EfficientNet-B0 under identical data conditions for a genuinely fair comparison
- Build a challenging ID test set consisting only of the hardest-to-detect fake samples that both models struggle with
- Evaluate across five practical dimensions — detection ability, robustness, FLOPs, model size, and inference speed



DATASET

- Used FaceForensics++ c23 with two manipulation types: DeepFakes (autoencoder-based) and FaceSwap (graphics-based) — 3,000 videos total
- Official split followed: 720 train / 140 val / 140 test per category, with 30 frames extracted per video giving ~90,000 total frames
- Faces detected using OpenCV Haar Cascade classifier, cropped with 30% margin and resized to 256×256 — frames with no face detected were discarded, yielding 12,458 clean test frames



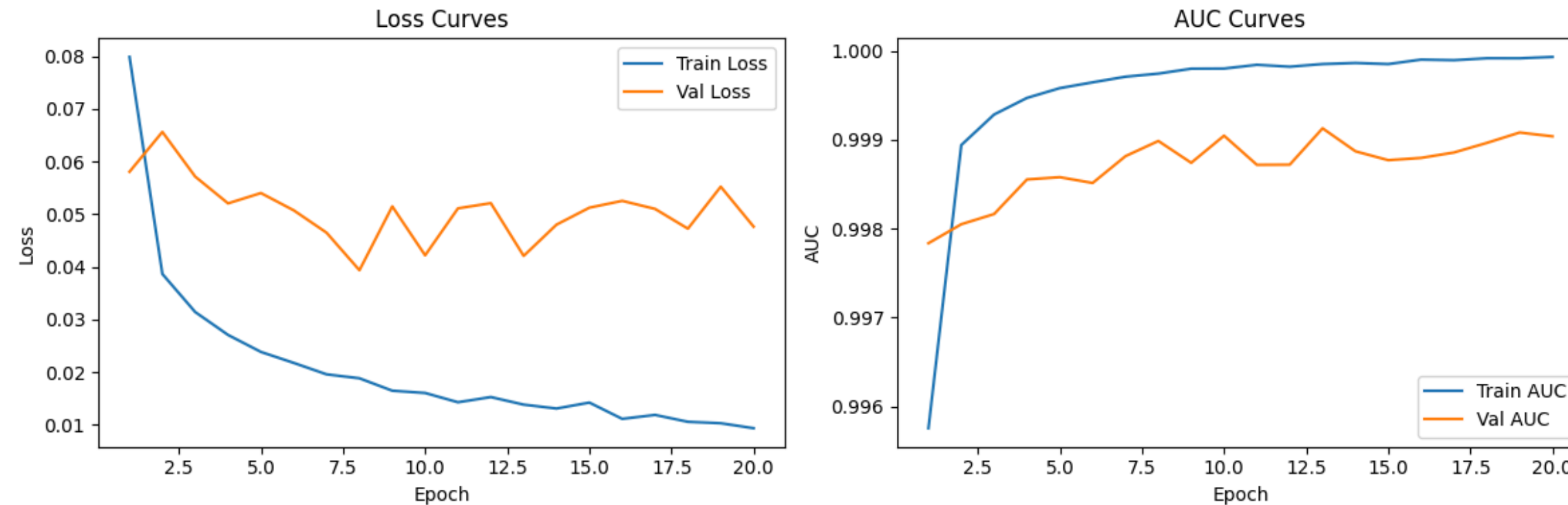
ID TEST SET

- Both trained models were run on all fake test frames; per-frame fake probability scores were averaged per video, and the bottom two-thirds of fake videos with the lowest predicted fake scores — those most confusing for both models — were retained and intersected to form the ID test set
- Final ID test set contains 8,021 frames across 272 videos — 4,045 real, 2,125 DeepFake, and 1,851 FaceSwap
- Five perturbed versions generated — JPEG, Blur, Contrast, Saturation, and Gaussian Noise — simulating real-world social media post-processing conditions

MODELS

- MesoNet-4 is a lightweight purpose-built deepfake detector with only 27,977 parameters — four convolutional blocks with dropout, trained from scratch
- EfficientNet-B0 is a pretrained ImageNet model fine-tuned for binary detection — compound-scaled architecture with 4 million parameters
- Both trained with Adam optimiser, BCELoss, batch size 32, for 20 epochs on identical data

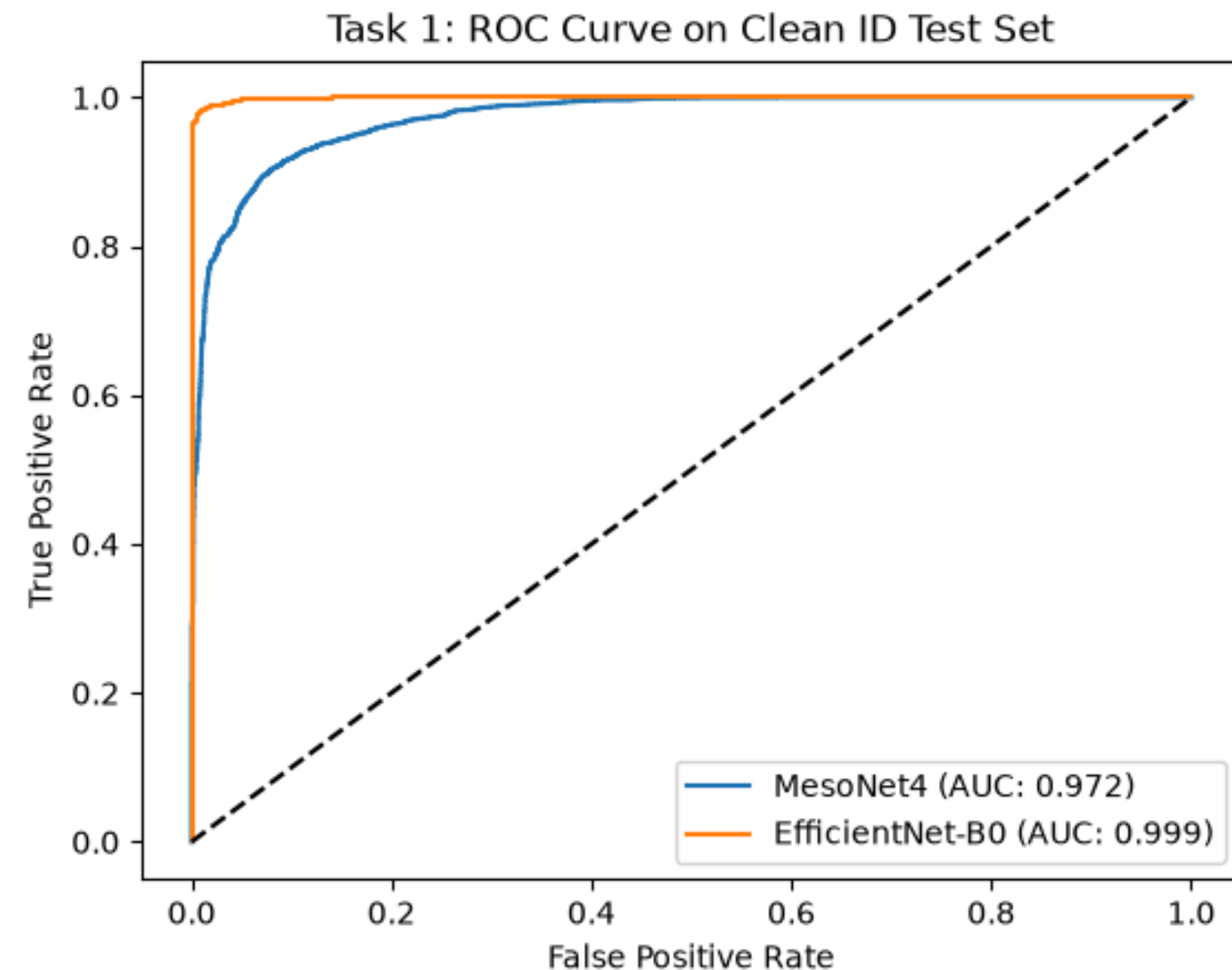
TRAINING



- MesoNet-4 converges within just 3 epochs, reaching a validation AUC of 0.9991 that remains stable for all 20 epochs
- A small gap between training loss (0.010) and validation loss (0.049) indicates mild overfitting, well controlled by 50% dropout in the classifier

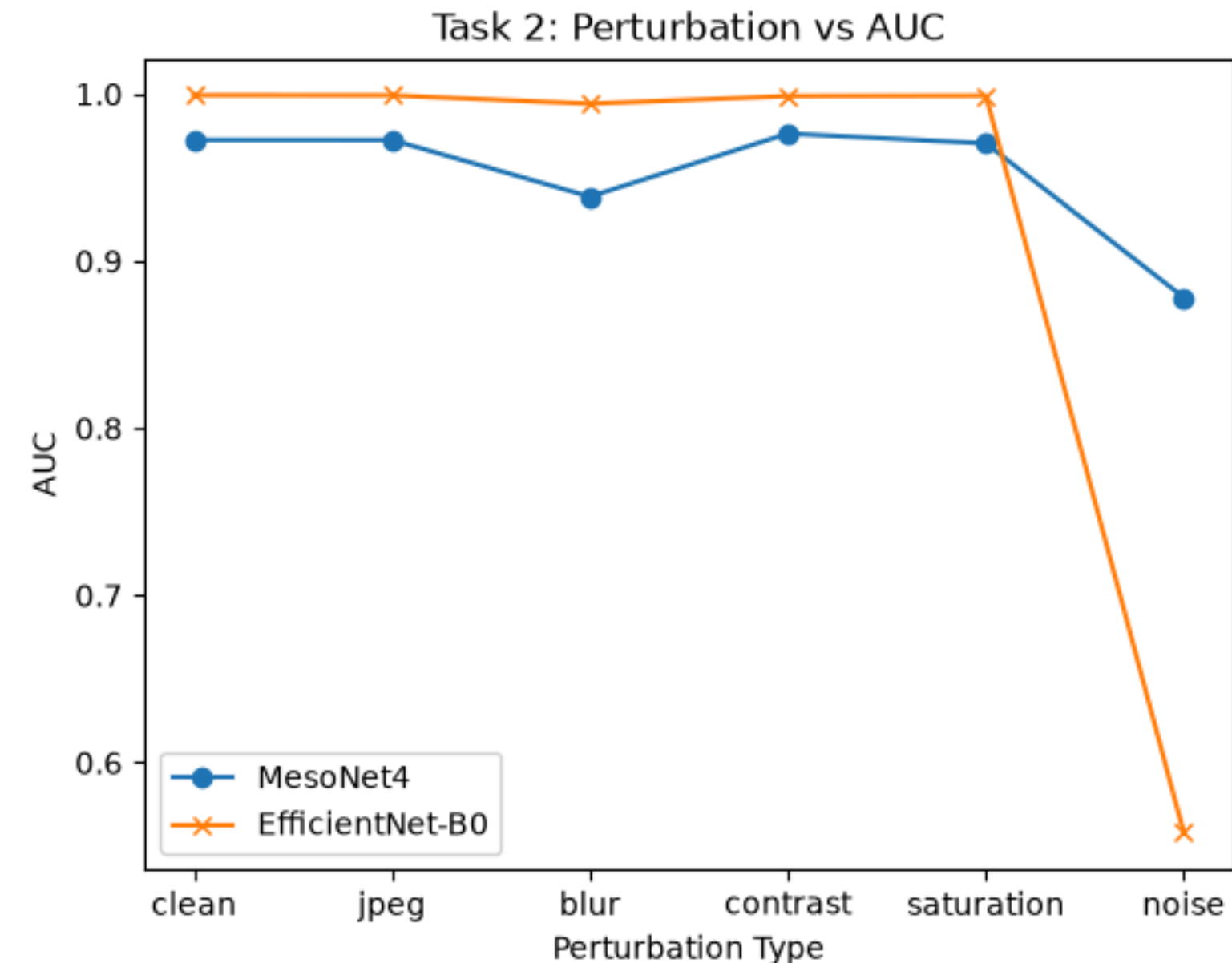
DETECTION ABILITY

- On the standard test set, MesoNet-4 achieves $AUC = 0.9972$ and EfficientNet-B0 achieves a perfect $AUC = 1.000$
- On the harder ID test set, EfficientNet-B0 leads at 0.999 vs MesoNet-4 at 0.972 — both strong results given MesoNet-4 uses 143× fewer parameters

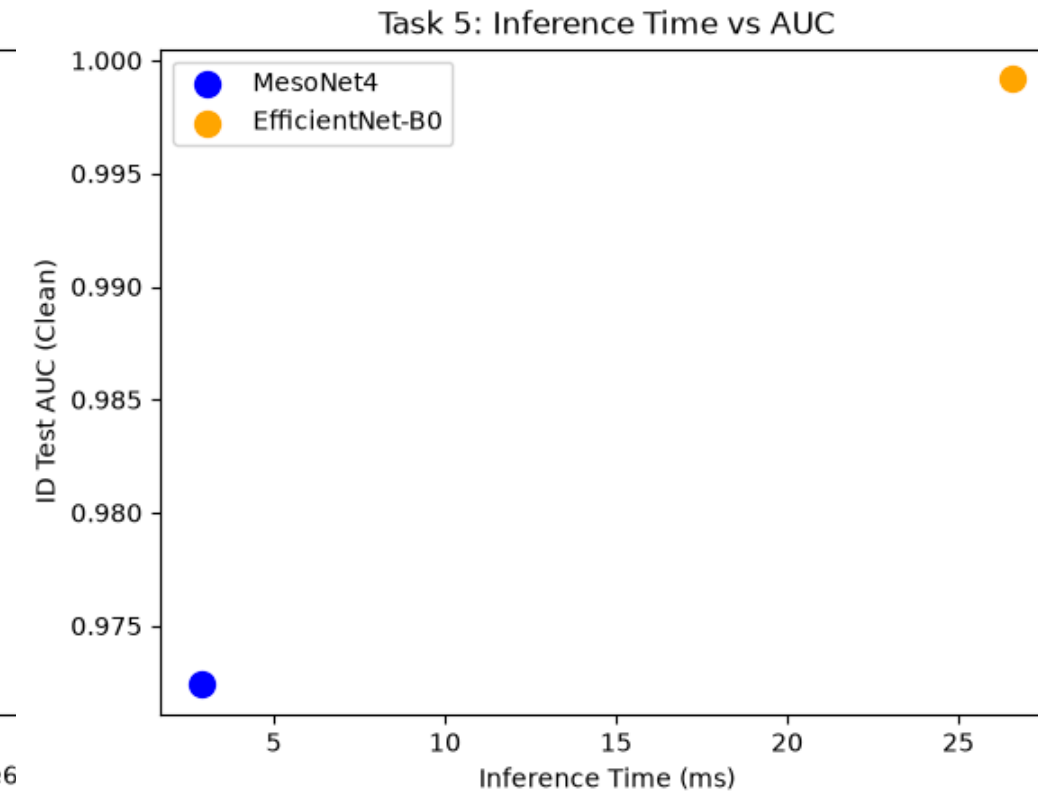
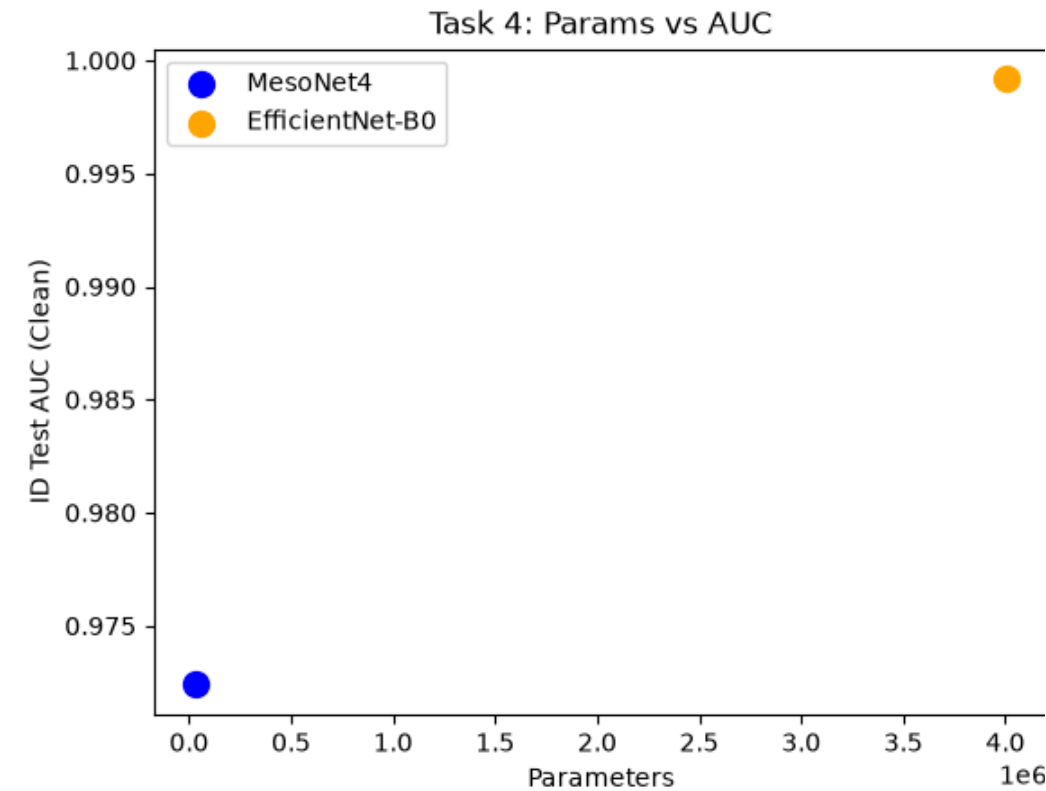
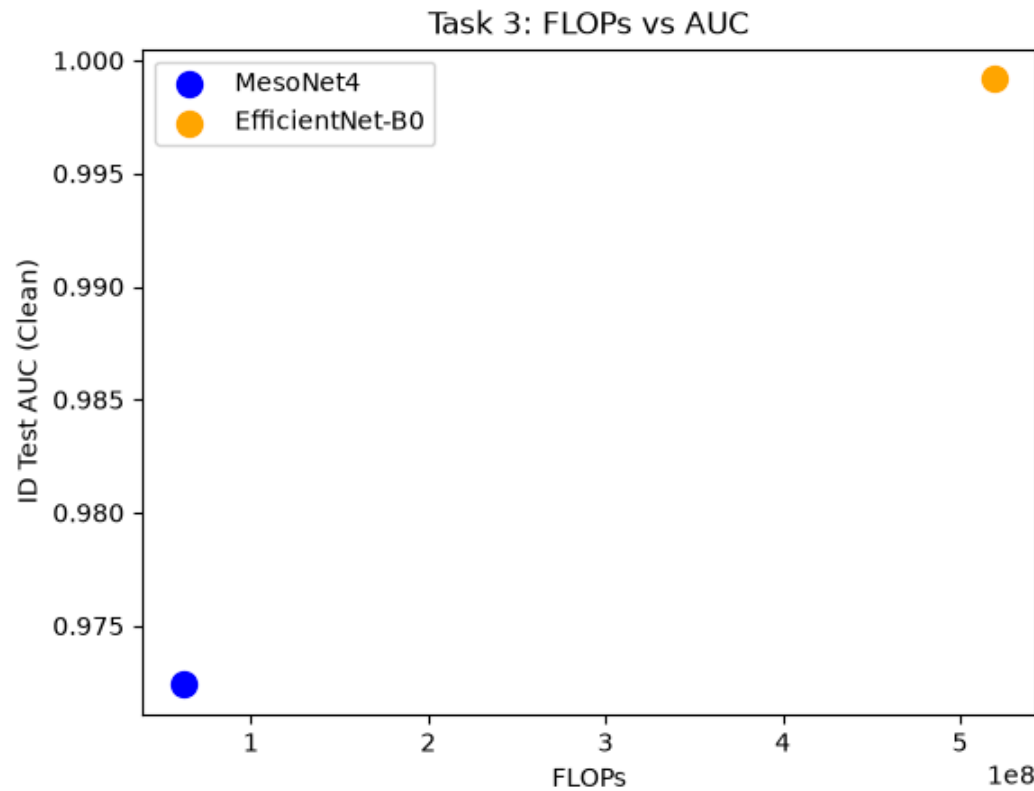


ROBUSTNESS UNDER PERTURBATION

- Both models handle JPEG, blur, contrast and saturation perturbations well — performance stays near clean levels
- Under Gaussian noise, EfficientNet-B0 catastrophically collapses from 0.999 to 0.555 near random chance while MesoNet-4 holds at 0.878



EFFICIENCY



- MesoNet-4 uses 8.4× fewer FLOPs (61.5M vs 519.7M) and has 143× fewer parameters (27,977 vs 4,008,829)
- MesoNet-4 runs at 2.86 ms per image versus 26.60 ms — 9.3× faster, leaving substantial headroom for real-time deployment

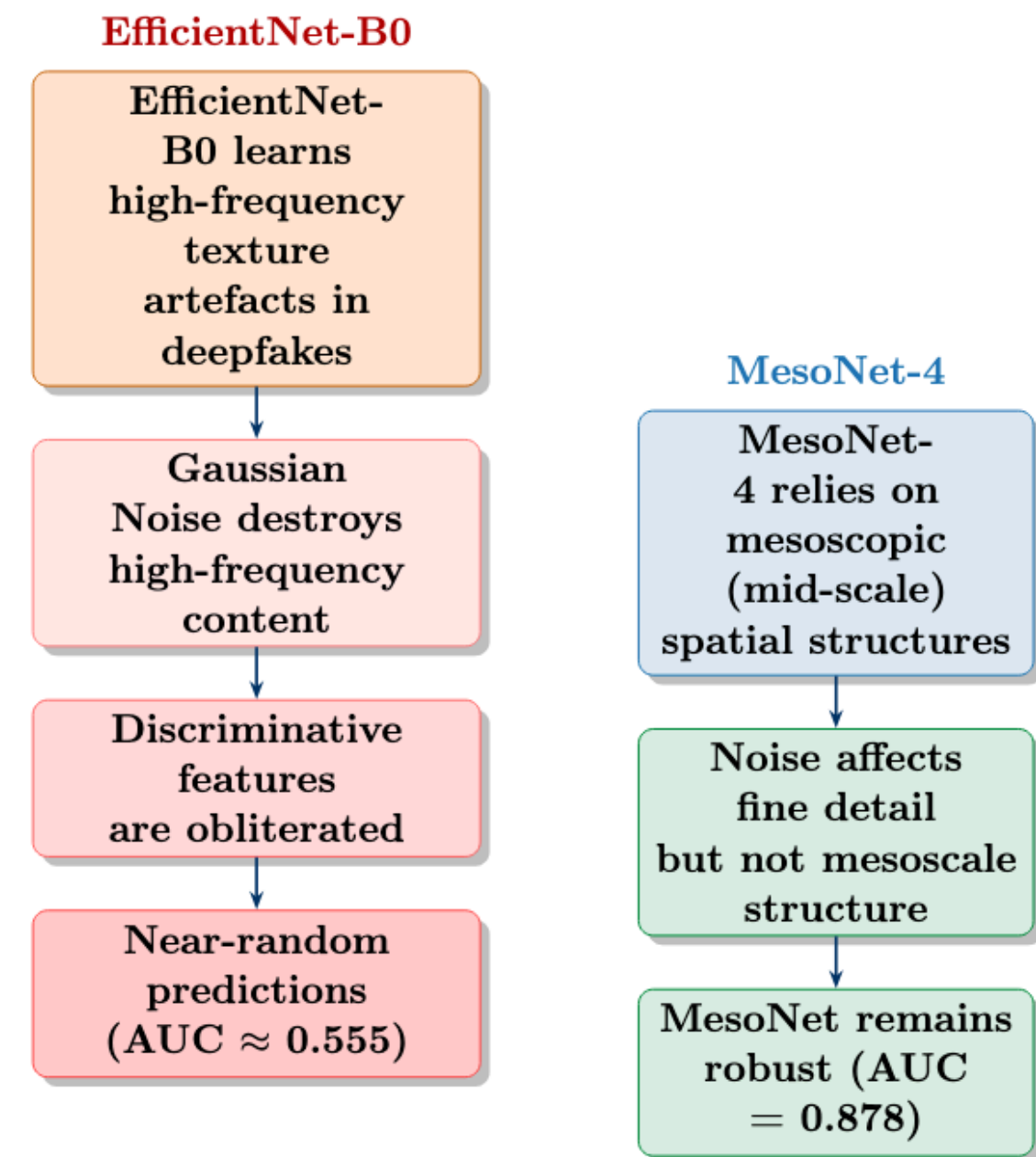
HEAD-TO-HEAD SUMMARY

- EfficientNet-B0 wins on raw accuracy in clean conditions; MesoNet-4 wins on robustness, speed, size and compute
- Overall score: MesoNet-4 wins 4 out of 6 metrics

Metric	MesoNet-4	EfficientNet-B0	Winner
Standard Test AUC	0.9972	1.0000	EfficientNet
ID Test AUC	0.9972	0.9990	EfficientNet
Noise Robustness	0.878	0.555	MesoNet
FLOPs	61.5M	519.7M	MesoNet
Parameters	27,977	4,008,829	MesoNet
Inference Speed	2.86 ms	26.60 ms	MesoNet
Overall Score	4/6	2/6	MesoNet wins

WHY DOES NOISE BREAK EFFICIENT NET?

- EfficientNet-B0 relies on fine high-frequency texture artefacts to detect fakes — Gaussian noise destroys exactly those features, causing predictions to collapse to near-random
- MesoNet-4 operates at a mesoscopic scale using broader mid-level spatial patterns that are far less sensitive to pixel-level noise — which is why it remains robust



CONCLUSION

- Both models are strong in-domain detectors, but EfficientNet-B0's accuracy advantage on clean data completely disappears under Gaussian noise
- MesoNet-4 is 8.4× more efficient, 143× smaller, and 9.3× faster — making it the more practical and robust real-world choice
- No single model is universally superior — the right choice depends entirely on the deployment environment and expected noise conditions

FUTURE WORK

- Extend evaluation to additional datasets such as Celeb-DF and DFDC to test cross-dataset generalisation
- Move to video-level temporal detection using CNN-LSTM or transformer architectures to capture inconsistencies across frames
- Train EfficientNet-B0 with Gaussian noise augmentation during training to close the robustness gap identified in this benchmark





**Thank
You**